

Microeconomics Analysis: Decision Theory II, Choice under Uncertainty

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Reading List

- D. Kreps “Microeconomic Foundations: Choice and Competitive Markets”, Chapter 5.
- A. Mas-Colell, M. Whinston and J.Green, “Microeconomic Theory”, Chapter 6.

6. Probability

In decisions under risk, the space of objects X can be seen as a set of probability distributions or random variables. Though a unified theory could be presented, a separate presentation of finite and continuous random variables help to deal with different models and economic frameworks

1 Basics of Probability

2 Lotteries

3 Expected Utility

4 Risk Aversion

Probability Space

A probability space is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

Sample space Ω . The set of all possible outcomes.

Events $\mathcal{F}$. A collection of subsets of Ω (the events) satisfying:

- $\Omega \in \mathcal{F}$,
- if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$,
- if $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Probability $\mathbb{P}$. A function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ such that

$$\mathbb{P}(\Omega) = 1, \quad \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

for disjoint events A_i .

Random Variable

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

Definition (Random Variable)

A random variable is a function

$$X : \Omega \rightarrow \mathbb{R}$$

such that for every $x \in \mathbb{R}$,

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$$

For a random variable X , the *cumulative distribution function* is defined by

$$F_X(x) = \mathbb{P}(X \leq x)$$

Note that the CDF fully describes the random variable, everything else (PDF, moments) is derived from CDF.

Support

Definition (Support)

Let X be a random variable. The support of X is

$$\text{supp}(X) = \{x \in \mathbb{R} : \mathbb{P}(X \in U) > 0 \text{ for every open set } U \text{ containing } x\}.$$

Equivalently, it is the smallest closed set $S \subseteq \mathbb{R}$ such that $\mathbb{P}(X \in S) = 1$.

Support is essentially a set of “possible” realisations of the random variable. Easy to define for discrete variables, harder for continuous.

Support: Examples

Discrete case

If X takes values in a countable set $\{x_1, x_2, \dots\}$, then

$$\text{supp}(X) = \{x_i : \mathbb{P}(X = x_i) > 0\}$$

The distribution is described by the probabilities

$$p_i = \mathbb{P}(X = x_i), \quad \sum_i p_i = 1$$

Continuous case

If X has density f , then

$$F_X(x) = \int_{-\infty}^x f(t) dt, \quad F_X(x) = \int_{\text{supp}(X)} f(t) dt = 1$$

The support is (essentially) the set

$$\{x \in \mathbb{R} : f(x) > 0\}$$

Finite Support: Expectation and Variance

Let $X = \{x_1, \dots, x_n\}$, be a random variable with finite support, and $p_i = \mathbb{P}(X = x_i)$, $\sum_{i=1}^n p_i = 1$.

Definition (Expected Value)

The expected value (or mean) of X is

$$\mathbb{E}[X] = \sum_{i=1}^n x_i p_i.$$

Definition (Variance)

The variance of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \sum_{i=1}^n (x_i - \mathbb{E}[X])^2 p_i.$$

Equivalently,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Continuous Case: Expectation and Variance

Let X be a continuous random variable with density f .

Definition (Expected Value)

The expected value (or mean) of X is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x) dx,$$

provided the integral $\int_{-\infty}^{\infty} |x|f(x) dx$ is finite.

Definition (Variance)

If $\mathbb{E}[X]$ exists, the variance of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \int_{-\infty}^{\infty} (x - \mathbb{E}[X])^2 f(x) dx.$$

Example: A Random Variable Without Expectation

Cauchy Distribution

Define

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan(x), \quad x \in \mathbb{R}.$$

Then F is increasing,

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow \infty} F(x) = 1, \quad f(x) = F'(x) = \frac{1}{\pi} \frac{1}{1+x^2}.$$

Consider the expectation:

$$\int_{-\infty}^{\infty} |x| f(x) dx = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{|x|}{1+x^2} dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx = \frac{1}{\pi} \ln(1+x^2) \Big|_0^{\infty} = \infty$$

Hence $\mathbb{E}[X]$ does not exist.

Affine Transformations

Definition

For real constants a, b define the function

$$h : \mathbb{R} \rightarrow \mathbb{R}, \quad h(x) = a + bx,$$

and the transformed random variable $Y = h(X) = a + bX$.

Expectation and Variance

$$\mathbb{E}[Y] = \mathbb{E}[a + bX] = a + b \mathbb{E}[X],$$

$$\text{Var}(Y) = \text{Var}(a + bX) = b^2 \text{Var}(X).$$

Jensen's Inequality (Strictly Concave h)

Theorem

Let X be a random variable and let $h : \mathbb{R} \rightarrow \mathbb{R}$ be strictly concave. Then

$$\mathbb{E}[h(X)] \leq h(\mathbb{E}[X]),$$

with equality if and only if X is almost surely constant.

Two-point case.

Assume X takes exactly two values x_1, x_2 with $\mathbb{P}(X = x_1) = p, \mathbb{P}(X = x_2) = 1 - p, p \in [0, 1]$. By the definition of expectation,

$$\mathbb{E}[X] = px_1 + (1 - p)x_2, \quad \mathbb{E}[h(X)] = ph(x_1) + (1 - p)h(x_2).$$

Since h is concave, for any x_1, x_2 and any $p \in [0, 1]$,

$$h(px_1 + (1 - p)x_2) \geq ph(x_1) + (1 - p)h(x_2),$$

with inequality strict for $p \in (0, 1)$. Thus, $h(\mathbb{E}[X]) \geq \mathbb{E}[h(X)]$ with inequality strict for interior p . □

Finite Probability Space and Independence

Let $\Omega = \{\omega_1, \dots, \omega_n\}$ be a finite set (the set of possible states).

Assign numbers $p_i = \mathbb{P}(\{\omega_i\})$ such that

$$p_i \geq 0, \quad \sum_{i=1}^n p_i = 1.$$

For any event $E \subseteq \Omega$, probability of E is defined as

$$\mathbb{P}(E) = \sum_{\omega_i \in E} p_i.$$

Definition

Two events $E, F \subseteq \Omega$ are independent if

$$\mathbb{P}(E \cap F) = \mathbb{P}(E)\mathbb{P}(F).$$

Conditional Probability and Bayes' Formula

Definition

Let $F \subseteq \Omega$ with $\mathbb{P}(F) > 0$. The conditional probability of E given F is

$$\mathbb{P}(E|F) = \frac{\mathbb{P}(E \cap F)}{\mathbb{P}(F)}.$$

From the definition,

$$\mathbb{P}(E \cap F) = \mathbb{P}(E|F)\mathbb{P}(F) = \mathbb{P}(F|E)\mathbb{P}(E).$$

If $\mathbb{P}(F) > 0$, dividing both sides by $\mathbb{P}(F)$ gives

$$\mathbb{P}(E|F) = \frac{\mathbb{P}(F|E)\mathbb{P}(E)}{\mathbb{P}(F)}$$

If $\{E_1, \dots, E_k\}$ is a partition of Ω , then

$$\mathbb{P}(E_j|F) = \frac{\mathbb{P}(F|E_j)\mathbb{P}(E_j)}{\sum_{i=1}^k \mathbb{P}(F|E_i)\mathbb{P}(E_i)}.$$

Stochastic Dominance

Let F and G be cumulative distribution functions.

Definition (First-Order Stochastic Dominance (FOSD))

F first-order stochastically dominates G if

$$F(x) \leq G(x) \quad \text{for all } x,$$

with strict inequality for some x .

Definition (Second-Order Stochastic Dominance (SOSD))

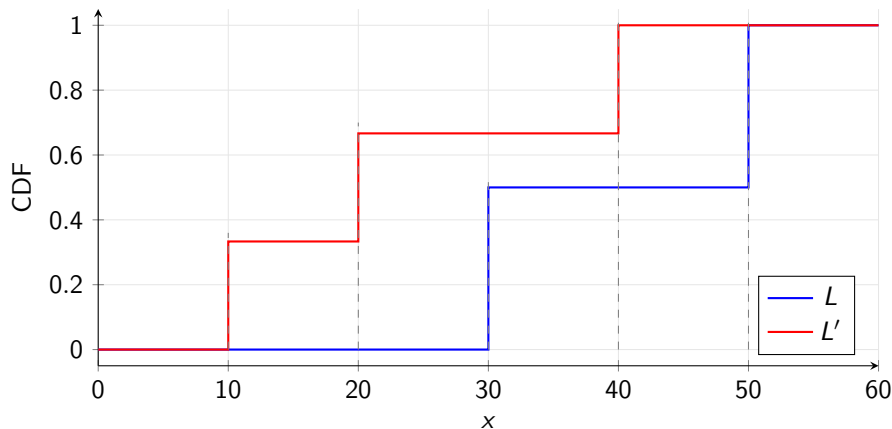
F second-order stochastically dominates G if

$$\int_{-\infty}^x F(t) dt \leq \int_{-\infty}^x G(t) dt \quad \text{for all } x,$$

with strict inequality for some x .

FOSD: Discrete Distributions

Consider two lotteries: $L = [\frac{1}{2}, \frac{1}{2}; 30, 50]$ and $L' = [\frac{1}{3}; \frac{1}{3}; \frac{1}{3}; 10, 20, 40]$.

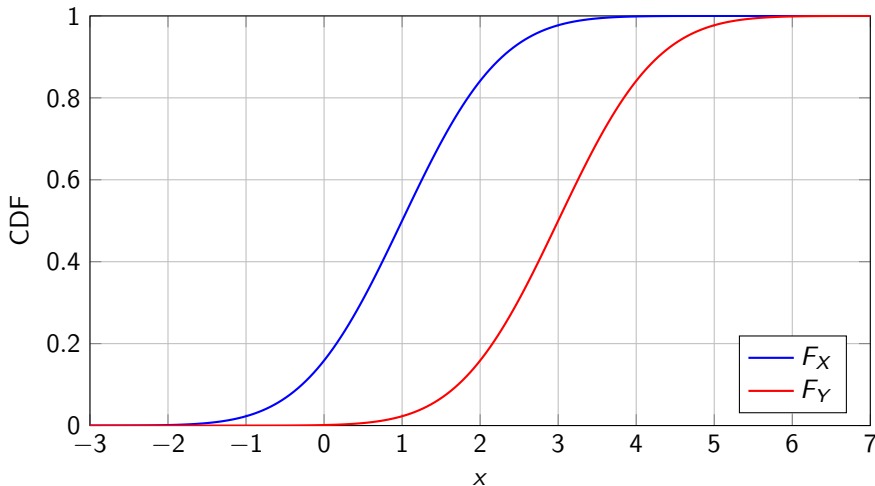


Thus, $L \succ_{FOSD} L'$.

FOSD: Continuous Distribution

Let

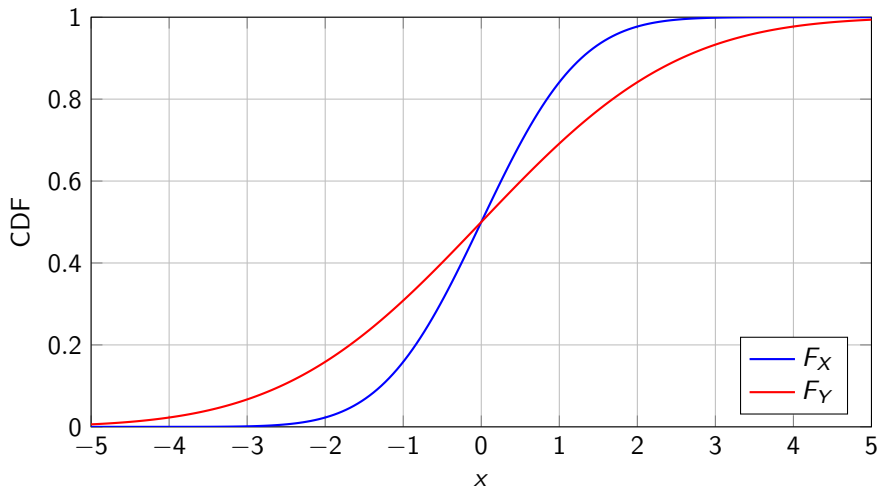
$$X \sim \mathcal{N}(1, 1), \quad Y \sim \mathcal{N}(3, 1).$$



Since $F_Y(x) \leq F_X(x)$ for all x , $Y \succ_{FOSD} X$.

SOSD

$$X \sim \mathcal{N}(0, 1), \quad Y \sim \mathcal{N}(0, 2).$$



$$\forall x \quad \int_{-\infty}^x F_X(t) dt \leq \int_{-\infty}^x F_Y(t) dt \quad \Leftrightarrow \quad X \succ_{SOSD} Y$$

FOSD implies SOSD

Theorem

If $F \succ_{\text{FOSD}} G$, then $F \succ_{\text{SOSD}} G$.

Proof. FOSD means

$$F(x) \leq G(x) \quad \text{for all } x \in \mathbb{R}.$$

Integrate both sides from $-\infty$ to x :

$$\int_{-\infty}^x F(t) dt \leq \int_{-\infty}^x G(t) dt.$$

Rearranging,

$$\int_{-\infty}^x (G(t) - F(t)) dt \geq 0 \quad \text{for all } x.$$

Since both F and G are CDFs,

$$\int_{-\infty}^{\infty} (G(t) - F(t)) dt = 0.$$

Therefore,

$$F \succ_{\text{SOSD}} G. \quad \square$$

FOSD implies larger expectation for any increasing u

Theorem

If $F \succ_{\text{FOSD}} G$, then for every increasing function $u : \mathbb{R} \rightarrow \mathbb{R}$

$$\mathbb{E}_F[u(X)] \geq \mathbb{E}_G[u(X)].$$

Proof. Write expectations:

$$\mathbb{E}_F[u(X)] = \int_{-\infty}^{\infty} u(x) dF(x), \quad \mathbb{E}_G[u(X)] = \int_{-\infty}^{\infty} u(x) dG(x).$$

Apply integration by parts:

$$\int_{-\infty}^{\infty} u(x) dH(x) = [u(x)H(x)]_{x=-\infty}^{x=+\infty} - \int_{-\infty}^{\infty} H(x) du(x).$$

Because $H(-\infty) = 0$ and $H(+\infty) = 1$, applying to F and G :

$$\mathbb{E}_F[u] - \mathbb{E}_G[u] = (u(+\infty) - \int_{-\infty}^{\infty} F(x) du(x)) - (u(+\infty) - \int_{-\infty}^{\infty} G(x) du(x)) = \int_{-\infty}^{\infty} (G(x) - F(x)) du(x) > 0$$

SOSD implies larger expectation for increasing concave u

Theorem

If $F \succ_{\text{SOSD}} G$, then for every increasing concave function $u : \mathbb{R} \rightarrow \mathbb{R}$

$$\mathbb{E}_F[u(X)] \geq \mathbb{E}_G[u(X)].$$

Proof. Write the difference in expectations:

$$\mathbb{E}_F[u(X)] - \mathbb{E}_G[u(X)] = \int_{-\infty}^{\infty} u(x) dF(x) - \int_{-\infty}^{\infty} u(x) dG(x) = \int_{-\infty}^{\infty} (G(x) - F(x)) du(x).$$

Define

$$A(x) := \int_{-\infty}^x (G(t) - F(t)) dt.$$

Then $A'(x) = G(x) - F(x)$ and, under SOSD, $A(x) \geq 0$ for all x . Thus,

$$\mathbb{E}_F[u] - \mathbb{E}_G[u] = \int_{-\infty}^{\infty} A'(x) u'(x) dx = [A(x) u'(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} A(x) u''(x) dx.$$

Because $A(-\infty) = A(+\infty) = 0$, the boundary term vanishes, hence

$$\mathbb{E}_F[u] - \mathbb{E}_G[u] = - \int_{-\infty}^{\infty} \underbrace{A(x)}_{\geq 0} \underbrace{u''(x)}_{< 0} dx \geq 0$$

Likelihood Ratio Dominance: Intuition

Start with uniform distribution over three outcomes $l < m < h$, that is,
 $[p_l, p_m, p_h] = [0.3, 0.3, 0.3]$.

$[0, 0.6, 0.4]$ is an FOSD improvement: the probability of the worst outcome is lower and the probabilities of the other two are higher

$[q_l, q_m, q_h] = [0, 0.4, 0.6]$ is also FOSD improvement, moreover, the probabilities of better outcomes are increased in greater proportions

$$\frac{q_l}{p_l} < \frac{q_m}{p_m} < \frac{q_h}{p_h}$$

These are called **likelihood ratios**, and when they are increasing in outcome x , the second lottery **LR dominates** the first one

Likelihood Ratio Dominance

Definition

Let $X = \{x_1 < x_2 < \dots < x_n\}$ be a finite ordered set. Let $p_i = \Pr_F(x_i)$ and $q_i = \Pr_G(x_i)$, with $p_i, q_i > 0$ for all i . We say that F likelihood-ratio dominates G (denoted $F \succ_{\text{LR}} G$) if the sequence

$$\frac{p_i}{q_i}$$

is weakly increasing in i , i.e.

$$\frac{p_{i+1}}{q_{i+1}} \geq \frac{p_i}{q_i} \quad \text{for all } i = 1, \dots, n-1.$$

Definition

Let F and G be two distributions with densities (or probability mass functions) f and g with common support S . We say that F likelihood-ratio dominates G (denoted $F \succ_{\text{LR}} G$) if the likelihood ratio

$$\frac{f(x)}{g(x)}$$

is weakly increasing in x on S .

LRD and FOSD

Theorem

Let F, G have densities $f, g > 0$ on a common support. If $\lambda(x) = \frac{f(x)}{g(x)}$ is increasing, then $F(x) \leq G(x)$ for all x .

Proof. Since

$$\int \lambda(x)g(x) dx = \int f(x) dx = 1,$$

the increasing function λ can cross the level 1 at most once.

Hence there exists c such that

$$\lambda(x) \leq 1 \text{ for } x \leq c, \quad \lambda(x) \geq 1 \text{ for } x \geq c.$$

Equivalently,

$$f(x) \leq g(x) \text{ for } x \leq c, \quad f(x) \geq g(x) \text{ for } x \geq c.$$

Thus

$$F(x) - G(x) = \int_{-\infty}^x (f(t) - g(t))dt \leq 0 \quad \text{for all } x,$$

because the integrand is nonpositive before c and any positive mass after c is needed to keep total mass equal. Therefore $F(x) \leq G(x)$ for all x .

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Simple Lotteries

We start building up from rational preferences $\succsim$ over set X —we will refer to it as a set space, or a set of outcomes.

Note that the rationality implies that these preferences are representable.

We can now ask a question whether we can extend our preferences over the distributions over state space:

Definition (Simple lottery)

A simple lottery is a probability distribution over a fixed set of outcomes $X = \{x_1, \dots, x_n\}$:

$$L = [p_1, \dots, p_n; x_1, \dots, x_n]$$

with $p_i \geq 0$ and $\sum_{i=1}^n p_i = 1$

This definition naturally extends to larger sets of outcomes.

Simple Lotteries

Definition (Compound lottery)

Given a set of lotteries $L_1, L_2, \dots, L_K$ and probabilities $q_1, \dots, q_K$, $\sum_{k=1}^K q_k = 1$, a compound lottery

$$L = [q_1, \dots, q_K; L_1, \dots, L_K]$$

is the risky alternative that yields lottery L_k with probability q_k .

Thus, a compound lottery is a probability distribution over the lotteries, which can be reduced to a lottery over the set of outcomes.

Assumption

Decision makers consider compound lotteries in their reduced form.

Lottery Reduction

Simple Lottery

Ann owns goods at home worth £4M, overseas £12M, the latter perishes at sea with 25% chance; her final wealth is the lottery

$$A = \left[\frac{1}{4}, \frac{3}{4}; 4, 16 \right]$$

Compound Lottery

Bob owns £4M at home, £12M overseas, but the latter is confiscated with 50% chance before it is shipped, in which case it still perishes at sea with 25% chance; his final wealth is lottery

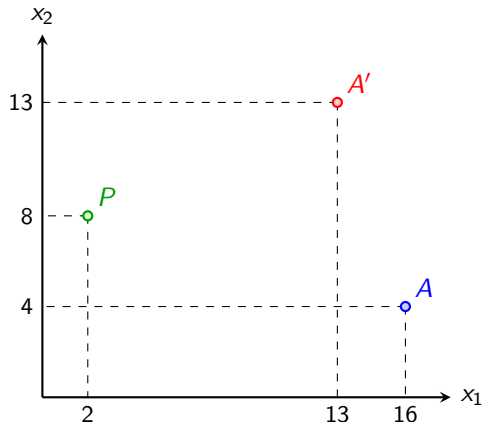
$$B = \left[\frac{1}{2}, \frac{1}{2}; 4, A \right]$$

$$B = \left[\frac{1}{2}, \frac{1}{2}; 4, A \right] = \left[\frac{5}{8}, \frac{3}{8}; 4, 16 \right]$$

Plotting lotteries with fixed probabilities

When probabilities are fixed, we can consider the space of lottery outcomes to represent the lottery.

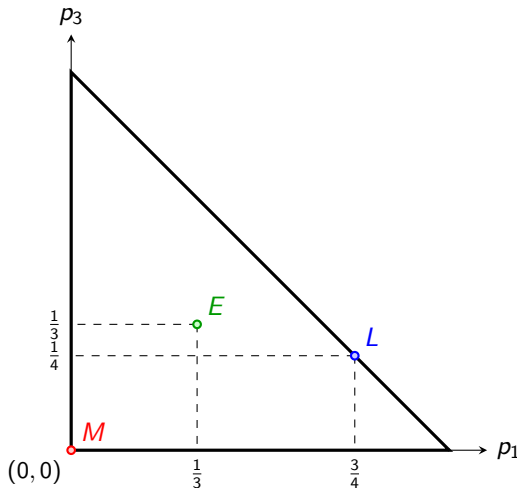
- A indicates Ann's initial lottery
- A' is the lottery when Ann sells the goods for £9M and wires money home
- P is the lottery of the pirate with initial wealth of £2M who was hunting Ann's shipment, but recovers only half of it's value if succeeds.
- Note, that we cannot plot Bob here: his probabilities are different.



Plotting lotteries with fixed prizes

This illustration is referred to as a Marschak-Machina triangle. Let $x_3 \succ x_2 \succ x_1$ and $L = [p_1, p_2, p_3; x_1, x_2, x_3]$

- Lottery L puts probability $3/4$ on the worst outcome and probability $1/4$ on the best outcome.
- Lottery M delivers the median outcome for sure.
- Lottery E puts equal probability on each outcome.



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Preferences over Lotteries

Consider a space of simple lotteries over $X = \{x_1, \dots, x_n\}$:

$$\mathcal{L} = \{L = [p_1, \dots, p_n; x_1, \dots, x_n]\}$$

We know that preferences are representable if they are rational and continuous:

- **complete:** $\forall L, L' \in \mathcal{L}$ we have either $L \succsim L'$ or $L' \succsim L$ or both
- **transitive:** if $L \succsim L'$ and $L' \succsim L''$ then $L \succsim L''$
- **continuous:** if $L \succsim L' \succsim L''$ then there exists probability p such that

$$pL + (1 - p)L'' \equiv [p, 1 - p; L, L''] \sim L'$$

Then, by Debreu's Theorem there is a function U such that $L \succsim L'$ if and only if $U(L) \geq U(L')$.

However, we cannot say much about the structure of $u(\cdot)$ at this point. In order to make progress, we need another assumption...

Independence Axiom

Axiom (Independence Axiom)

The preference relation on the space of simple lotteries satisfies the **independence axiom** if for all $L, L', L'' \in \mathcal{L}$ and $p \in (0, 1)$

$$L \succsim L' \quad \text{if and only if} \quad pL + (1 - p)L'' \succsim pL' + (1 - p)L''$$

Intuition: when ranking L and L' the decision maker receives the same if the coin flip ends up Tails, so the only thing that matters for them is how they rank what they get when the outcome is Heads

The axiom expresses that the decision maker does not worry about what “could have happened but did not”

Expected Utility

Definition (Expected Utility)

The utility function $U : \mathcal{L} \rightarrow \mathbb{R}$ has the (von Neumann-Morgenstern) expected utility form if there exist numbers $u(x_i)$ such that for every $L = [p_1, \dots, p_n; x_1, \dots, x_n] \in \mathcal{L}$

$$U(L) = \sum p_i u(x_i)$$

Note, that $U([1, x_i]) = u(x_i)$, that is, the value of the “lottery” which gives x_i for sure equals to the utility of x_i .

An increasing utility function $u(x_i)$ defined on monetary payoffs is referred to as **Bernoulli utility function**

Linearity of Expected Utility

Proposition

If $U : \mathcal{L} \rightarrow \mathbb{R}$ has the expected utility form if and only if it is linear:

$$U \left(\sum_{k=1}^K q_k L_k \right) = \sum_{k=1}^K q_k U(L_k)$$

for any $L_k \in \mathcal{L}$ and $\sum_{k=1}^K q_k = 1$.

Proof. Suppose that U has the expected utility form, then

$$U \left(\sum_{k=1}^K q_k L_k \right) = \sum_{i=1}^n u(x_i) \left(\sum_{k=1}^K q_k p_i^k \right) = \sum_{k=1}^K q_k \left(\sum_{i=1}^n p_i^k u(x_i) \right) = \sum_{k=1}^K q_k U(L_k)$$

Suppose that U is linear. Then, any lottery $L = [p_1, \dots, p_n; x_1, \dots, x_n]$ can be written as a sum of degenerate lotteries $L^i = [1; x_i]$, hence

$$U(L) = U \left(\sum_{i=1}^n p_i L^i \right) = \sum_{i=1}^n p_i u(x_i)$$

hence U has the expected utility form.

Invariance to Affine Transformations

Theorem

Suppose that $U : \mathcal{L} \rightarrow \mathbb{R}$ is the expected utility function. Then V is another expected utility function if and only if there exist $a > 0$ and b such that

$$V(L) = aU + b$$

Check MWG Proposition 6.B.2 for the proof.

Note, that unlike usual utility, expected utility function is invariant to linear transformations, not any monotone transformations!

This is important, because applying a non-linear transformation changes the risk attitude.

Expected Utility Theorem

Theorem (Expected Utility Theorem)

Suppose that agent's preferences over lotteries $\mathcal{L}$ are complete, transitive, continuous and satisfy the Independence Axiom. Then, there exists a function $u(x)$ such that the utility of any lottery $L = [p_1, \dots, p_n; x_1, \dots, x_n] \in \mathcal{L}$ can be written as

$$U(L) = p_1 u(x_1) + \dots + p_n u(x_n)$$

Formal treatment can be found in Kreps, Proposition 5.4

The key is that by the Independence Axiom the utility of lottery L can be decomposed as the sum of utilities received in each mutually exclusive event when a particular outcome occurs.

We are going to do a geometric proof for three outcomes using Marschak-Machina triangle

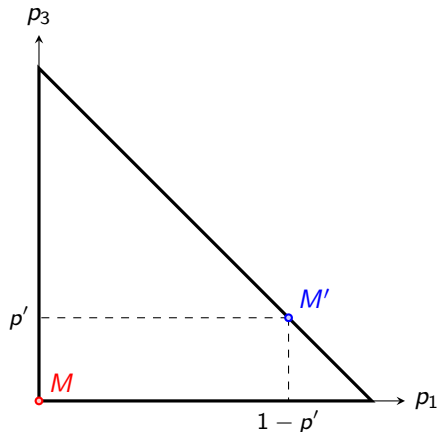
Proof of the Expected Utility Theorem I

- We start with three fixed outcomes $x_3 \succ x_2 \succ x_1$
- Assign

$$U([1, x_1]) = u(x_1) = 0$$

$$U([1, x_3]) = u(x_3) = 1$$

- How can we assign utility to $M = [1; x_2]$?
- Use continuity: there must be $M' = [1 - p', 0, p'; x_1, x_2, x_3]$ such that $M \sim M'$
- Then we can assign $U(M) = u(x_2) = p'$.



Proof of the Expected Utility Theorem II

- Now because $M \sim M'$ by Independence Axiom we must have that for any $q \in [0, 1]$

$$M \sim qM' + (1 - q)M$$

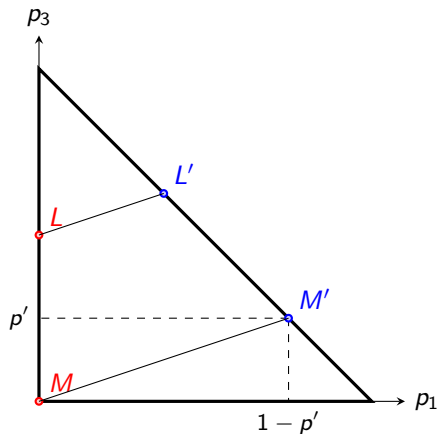
- Hence we have our first indifference curve!
- Now fix some α and consider

$$L = \alpha M + (1 - \alpha)[1; x_3]$$

$$L = \alpha M' + (1 - \alpha)[1; x_3]$$

- Again, applying the Independence Axiom to L and L' we get a parallel indifference curve with slope $p'/(1 - p')$.
- Hence $\forall L \in \mathcal{L}$ we have $U(L) = a + bp_1 + cp_3$
- Checking corners: $a = p'$, $b = -p'$, $c = 1 - p'$.

$$U(L) = 0 \cdot p_1 + (1 - p_1 - p_3)p' + p_3 = p_1u(x_1) + p_2u(x_2) + p_3u(x_3)$$



EU and FOSD

Theorem

Let L and L' be lotteries with CDFs F and G . If $F \succ_{\text{FOSD}} G$ (i.e. $F(x) \leq G(x)$ for all x), then any expected-utility maximiser with an increasing utility $u : \mathbb{R} \rightarrow \mathbb{R}$ (for which the expectations below exist) prefers L to L' :

$$\mathbb{E}_F[u(X)] \geq \mathbb{E}_G[u(X)].$$

Proof. As $u(x)$ is an increasing function, the result follows from our discussion of FOSD.

Note, that one structure of the state space almost doesn't matter: we just need to be able to re-arrange the outcomes so that u is increasing on the outcome space and then check the FOSD property.

Expected Utility: Discussion

Expected utility theorem explicitly relies on the continuity and the Independence Axiom.

Continuity implies the existence of indifference curves.

Independence Axiom implies that the indifference curves, when drawn in probability space, are:

- ① linear
- ② parallel

This provides testable predictions for the Expected Utility Theory.

Allais Paradox

Maurice Allais (Nobel Prize 1988) proposed experiments producing consistent violations of the EU hypothesis

Allais paradox setup: choose either A or B and then either C or D:

(A) win \$1 million for sure

(B) win \$5 million with 10%, \$1 million with 89%, nothing with 1%

(C) win \$1 million with 11% and nothing with 89%

(D) win \$5 million with 10% and nothing with 90%

① What would you choose between A and B?

② What would you choose between C and D?

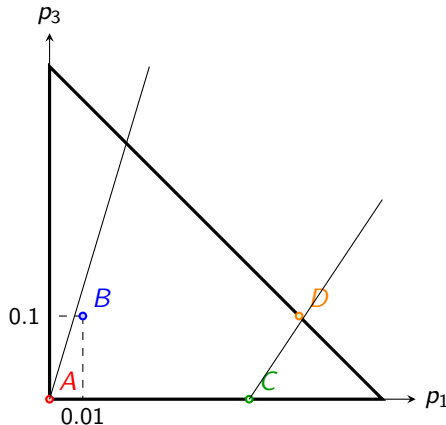
For most of people $A \succ B$, $D \succ C$ (at least when \$1m was big money)

We have three outcomes: $\{0, \$1, \$5\}$, so can it using Marschak-Machina triangle.

Allais Paradox and the Independence Axiom

- (A) win \$1 million for sure
- (B) win \$5 million with 10%, \$1 million with 89%, nothing with 1%
- (C) win \$1 million with 11% and nothing with 89%
- (D) win \$5 million with 10% and nothing with 90%

The indifference curve such that $A \succ B$ must be steeper than the indifference curve implying that $D \succ C$.



Algebraically, if $u(\$5m) = 1$, $u(\$1m) = z$, $u(0) = 0$ we get

$$A \succ B \Leftrightarrow z > 0.1 + 0.89z \Leftrightarrow 0.11z > 0.1 \Leftrightarrow C \succ D$$

1 Basics of Probability

2 Lotteries

3 Expected Utility

4 Risk Aversion

The St. Petersburg Paradox

The St. Petersburg Paradox

Toss a fair coin until the first head appears. If the first head occurs on toss n , the payoff is 2^n . That is,

$$L = \left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots; w+1, w+2, w+4, \dots \right]$$

where w is the initial wealth. How much would you pay to play?

Expected monetary value:

$$\mathbb{E}[X] = w + \sum_{n=1}^{\infty} \frac{1}{2^n} 2^n = w + \sum_{n=1}^{\infty} 1 = \infty.$$

Paradox: infinite expected value, yet people are only willing to pay a modest amount to play.

Bernoulli (1738): Evaluate the lottery using expected utility rather than expected value. E.g, set $u(x) = \log_2 x$ and $w = 0$

$$U(L) = \mathbb{E}[u(X)] = \sum_{n=1}^{\infty} \frac{1}{2^n} \log_2(2^{n-1}) = \sum_{n=1}^{\infty} \frac{n-1}{2^n} = 1 \quad \Rightarrow \quad L \sim [1; 2]$$

This expresses risk aversion which we are going to define and study.

Risk Aversion

Suppose that the agent has EU preferences:

$$U(L) = \mathbb{E}[u(x)]$$

where $u(x)$ is Bernoulli utility function.

Definition (Risk Aversion)

An agent is **risk averse** if they prefer the expected value of a lottery over final wealth to the lottery itself, i.e. if for all $L \in \mathcal{L}$

$$[1; \mathbb{E}x] \succ L \quad \Longleftrightarrow \quad u(\mathbb{E}x) > \mathbb{E}[u(x)]$$

Recall our discussion of Jensen's inequality:

Proposition

The expected utility maximizer is risk-averse if and only if their u is concave.

Risk Aversion and SOSD

Proposition

Consider two lotteries L_F and L_G over monetary outcomes and suppose that $F \succ_{SOSD} G$. Then any risk averse expected utility maximizer has $L_F \succ L_G$.

Proof. We have shown that a risk averse expected utility maximizer must have concave Bernoulli utility function $u(x)$.

We have shown that if u is increasing and strictly concave and $F \succ_{SOSD} G$ then

$$\mathbb{E}_F[u(x)] > \mathbb{E}_G[u(x)] \quad \Longleftrightarrow \quad U(L_F) > U(L_G)$$

Thus, $L_F \succ L_G$.

Certainty Equivalent and Risk Premium

Definition (Certainty Equivalent)

The **certainty equivalent** of lottery L is the sure outcome that the agent finds exactly as good as the lottery: $[1; CE(L)] \sim L$, or equivalently

$$u(CE(L)) = \mathbb{E}[u(x)]$$

Definition (Risk Premium)

The **risk premium** of lottery L is the difference between the monetary equivalent and the certainty equivalent of L :

$$RP(L) = \mathbb{E}[x] - CE(L)$$

Existence of certainty equivalent

Theorem

Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and (weakly) increasing. Let X be a random final wealth such that $\mathbb{E}[u(X)]$ exists (finite). Then there exists a unique $CE \in \mathbb{R}$ with

$$u(CE) = \mathbb{E}[u(X)],$$

Proof. Let $m \equiv \mathbb{E}[u(X)]$. Since u is real-valued on $\mathbb{R}$, set

$$u_{\min} := \inf_{x \in \mathbb{R}} u(x), \quad u_{\max} := \sup_{x \in \mathbb{R}} u(x).$$

Because $u(X)$ takes values in $[u_{\min}, u_{\max}]$, its expectation satisfies $u_{\min} \leq m \leq u_{\max}$.

If $m = u_{\min}$ or $m = u_{\max}$ then set CE with $u(CE) = m$. Otherwise $m \in (u_{\min}, u_{\max})$.

The continuous image $u(\mathbb{R})$ is a (connected) interval containing $[u_{\min}, u_{\max}]$. By the Intermediate Value Theorem there exists $CE \in \mathbb{R}$ with $u(CE) = m$.

If u is strictly increasing then u is invertible, so the solution CE is unique.

Alternative Representations of Risk Aversion

Theorem

The following are equivalent definitions for an agent (with vNM preferences represented by a strictly increasing Bernoulli utility function u on final wealth) being risk averse:

- *agent prefers “EV(L) for sure” to L , i.e., $u(\mathbb{E}x) > \mathbb{E}[u(x)]$*
- *$RP(L) > 0$*
- *agent's Bernoulli utility $u(x)$ is strictly concave*

In most of our settings we refer to the agent as ‘risk averse’ because the risk attitudes do not change on the space of the lotteries $\mathcal{L}$.

This implies that u is globally concave.

Numerical Example

Recall Ann we met in previous lecture, whose final wealth is given by a lottery $A = [\frac{1}{4}, \frac{3}{4}; 4, 16]$

Suppose that her Bernoulli utility function is $u(x) = \sqrt{x}$

Her expected utility

$$U(A) = \frac{1}{4}\sqrt{4} + \frac{3}{4}\sqrt{16} = 3.5$$

Thus, her certainty equivalent is

$$CE(A) = u^{-1}(U(A)) = 3.5^2 = 12.25$$

The monetary equivalent of the lottery is

$$\mathbb{E}[X] = \frac{1}{4}4 + \frac{3}{4}16 = 13$$

The risk premium is

$$RP(A) = \mathbb{E}[X] - CE(A) = 0.75$$

Illustration of Utility Function

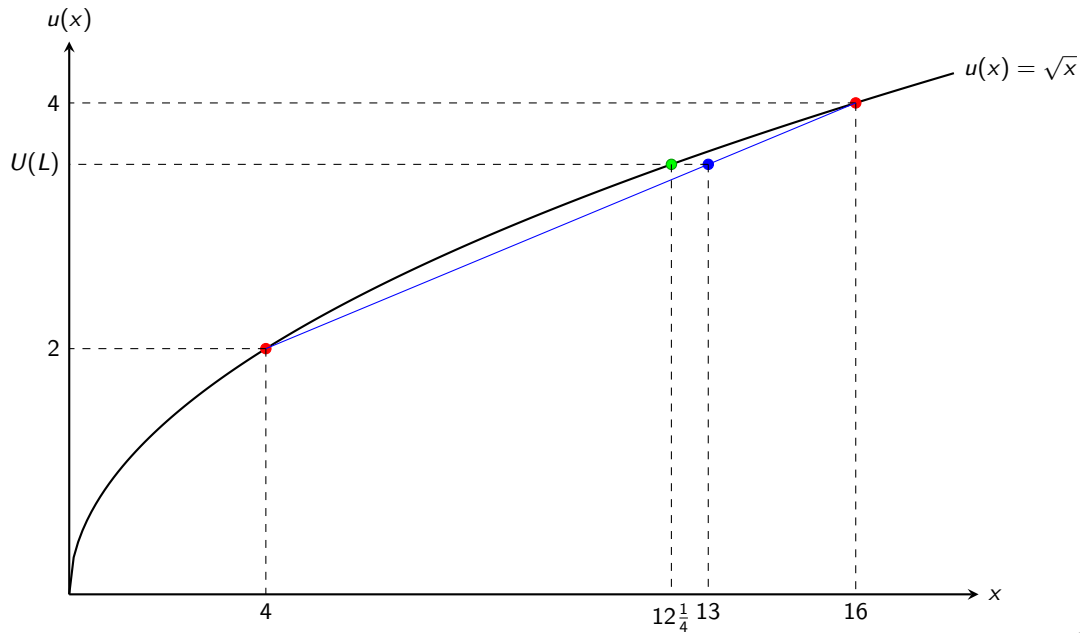


Illustration in State Space

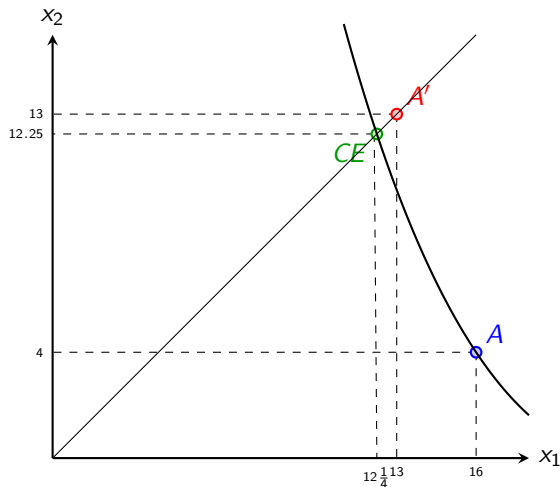
- Write out $U(A) = CE$ as

$$\frac{3}{4}\sqrt{x_1} + \frac{1}{4}\sqrt{x_2} = 3.5$$

- This gives

$$x_2 = (14 - 3\sqrt{x_1})^2$$

- $CE(A)$ is the intersection of the indifference curve with the 45-degree line.



Quantifying Risk Attitudes

How can we say that one agent is more risk-averse than another one?

Intuitively, if agent A's Bernoulli utility function is more concave than that of agent B, then agent A should be more risk-averse.

But just comparing u'' is a bad idea: if $v(x) = bu(x)$, then $v''(x) = bu''(x)$. So we may have $v'' > u''$, but they represent identical preferences!

Definition (Arrow-Pratt measure of absolute risk aversion)

$$A(x) = -\frac{u''(x)}{u'(x)}$$

- the second derivative of u normalized by the first derivative
- negative sign upfront makes $A(x)$ positive for concave $u(x)$
- invariant to affine transformations: $v(x) = a + bu(x)$ with $b > 0$ has literally the same $A(x)$ for all x

$$-\frac{v''(x)}{v'(x)} = -\frac{bu''(x)}{bu'(x)} = -\frac{u''(x)}{u'(x)}$$

recall our discussion of invariance with respect to affine transformations.

Arrow-Pratt Theorem

Theorem

The following statements are equivalent when comparing Bernoulli utility functions $u_1(x)$ and $u_2(x)$

- ① *an agent with utility function u_1 rejects any lottery rejected by an agent with Bernoulli utility u_2 (and the same initial wealth level)*
- ② *for every lottery, the risk premium of the lottery under Bernoulli utility function u_1 is larger than that under u_2*
- ③ *u_1 is more concave than u_2 , specifically, there is an increasing and strictly concave function g such that $u_1(x) = g(u_2(x))$ for all x*
- ④ *the Arrow-Pratt Absolute Measure of risk aversion of u_1 is, as a function of x , greater than that of u_2 , for all x*

All mean that u_1 is “more risk averse” than u_2

We prove equivalence between 3 and 4.

Proof of a Part of Arrow-Pratt Theorem

Proof. Differentiate $u_1(x) = g(u_2(x))$ twice:

$$u_1'(x) = g'(u_2(x)) u_2'(x),$$

$$u_1''(x) = g''(u_2(x)) (u_2'(x))^2 + g'(u_2(x)) u_2''(x).$$

Compute the Arrow-Pratt measure for u_1 :

$$A_1(x) = -\frac{u_1''(x)}{u_1'(x)} = -\frac{g''(u_2(x))(u_2'(x))^2 + g'(u_2(x))u_2''(x)}{g'(u_2(x))u_2'(x)} = -\frac{g''(u_2(x))}{g'(u_2(x))} u_2'(x) - \frac{u_2''(x)}{u_2'(x)} > A_2(x)$$

because $g' > 0$, $g'' < 0$ and $u_2' > 0$.

Constant Absolute Risk Aversion

CARA(a) has constant $A(x) = a$:

$$u(x) = -e^{-ax}$$

Check:

$$u'(x) = ae^{-ax}, \quad u''(x) = -a^2e^{-ax} \quad \Rightarrow \quad A(x) = -\frac{u''(x)}{u'(x)} = a$$

CARA(a) agent accepts the same gambles at every initial wealth level; they do not become more or less risk averse as their wealth increases

- think of it as a benchmark model
- higher a means that the agent is more risk-averse

The certainly equivalent for a CARA(a) agent of final wealth that is normally distributed with mean μ and variance σ^2 is $CE = \mu - \frac{1}{2}a\sigma^2$

- as if agent had “mean-variance” utility, popular in Finance

CARA with Normal Risk

Write out the expected utility:

$$U(L) = \mathbb{E}[-e^{-aX}] = - \int_{-\infty}^{\infty} e^{-ax} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx.$$

Combine exponents:

$$-ax - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{1}{2\sigma^2} \left[(x - (\mu - a\sigma^2))^2 - a^2\sigma^4 \right].$$

Hence

$$\mathbb{E}[e^{-aX}] = \exp\left(-a\mu + \frac{1}{2}a^2\sigma^2\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - (\mu - a\sigma^2))^2}{2\sigma^2}\right) dx.$$

The integral equals 1, so

$$U(L) = -\mathbb{E}[e^{-aX}] = -\exp\left(-a\mu + \frac{1}{2}a^2\sigma^2\right) = -e^{-aCE} \quad \Rightarrow \quad CE = \mu - \frac{1}{2}a\sigma^2.$$

Risk Premium under Expected Utility

Consider wealth x subject to additive risk:

$$X = x + \varepsilon, \quad \mathbb{E}[\varepsilon] = 0, \quad \text{Var}(\varepsilon) = \sigma^2.$$

Under expected utility using $CE = x - RP$:

$$u(x - RP) = \mathbb{E}[u(X)]$$

Second-order approximations around x :

$$u(x - RP) \approx u(x) - RP u'(x),$$

$$\mathbb{E}[u(X)] = \mathbb{E}[u(x + \varepsilon)] \approx u(x) + u'(x)\mathbb{E}[\varepsilon] + \frac{1}{2}u''(x)\mathbb{E}[\varepsilon^2].$$

Since $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{E}[\varepsilon^2] = \sigma^2$,

$$\mathbb{E}[u(X)] \approx u(x) + \frac{1}{2}u''(x)\sigma^2.$$

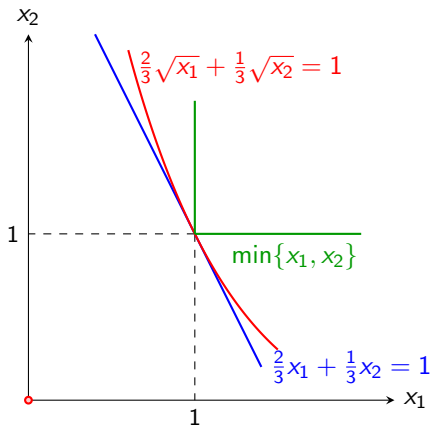
Equating both sides:

$$u(x) - RP u'(x) \approx u(x) + \frac{1}{2}u''(x)\sigma^2, \quad \Rightarrow \quad RP \approx -\frac{1}{2}\sigma^2 \frac{u''(x)}{u'(x)} = \frac{1}{2}\sigma^2 A(x),$$

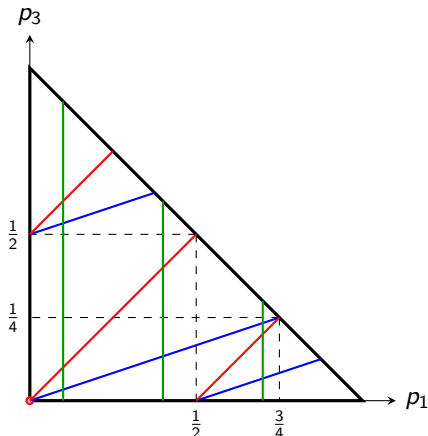
Graphical Representation of Risk Aversion

$$u(x) = x, u(x) = \sqrt{x}, \text{ absolute risk aversion}$$

$$\mathbb{P}(x_1) = 2/3$$



$$x_1 = 0, x_2 = 1, x_3 = 4$$



DARA

Consider $u(x) = \sqrt{x}$, then

$$A(x) = \frac{1}{2x}$$

which is decreasing in x .

Definition (DARA)

We say that preferences exhibit decreasing risk aversion if $A(x)$ is decreasing in x .

This is true: many people become less risk averse as their wealth goes up.

Perhaps it makes sense to think in relative terms: fraction of one's wealth...

Relative Risk Aversion

Definition (Arrow-Pratt measure of relative risk aversion)

$$R(x) = A(x)x = -\frac{u''(x)}{u'(x)}x$$

Special case: $A(x) = r/x$ (like in our $u = \sqrt{x}$ example)—constant relative risk aversion, $\text{CRRA}(r)$

Proposition

$\text{CRRA}(r)$ is $u = x^{1-r}$, $0 < r < 1$, and $u(x) = \ln x$, for $r = 1$

Verification

- former: $u'(x) = (1-r)x^{-r}$, $u''(x) = -r(1-r)x^{-r-1}$, hence $R(x) = -u''(x)x/u'(x) = r$
- latter: $u'(x) = 1/x$, $u''(x) = -1/x^2$, $R(x) = 1$.